Fiscal Year:	FY 2021	**Task Last Updated:**	FY 12/04/2020
PI Name:	Baker, John Ph.D.		
Project Title:	Defining the Relationship Between Simulated Weightlessness and Space Radiation on Cardiovascular Disease, and the Effectiveness of a Countermeasure		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	53226-3548	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-SRHHC. Appendix E: Space Radiobiology and Human Health Countermeasures Topics
Start Date:	02/04/2019	**End Date:**	02/03/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	December 2020 report: None.		
COI Name (Institution):	Kronenberg, Amy Sc.D. (Lawrence Berkeley National Laboratory) Willey, Jeffrey Ph.D. (Wake Forest University)		
Grant/Contract No.:	80NSSC19K0498		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal is one of two proposals selected for funding in response to solicitation NNJ16ZSA001N-SRHHC “Topic 6: Combined Effects of Simulated Weightlessness and Space Radiation on Cardiovascular Structure and Function,” and then descoped and merged at the direction of NASA with a second selected proposal (Principal Investigator-PI Michael Delp, PhD ; Radiation, Simulated Weightlessness and Countermeasures: Effects on Cerebral and Coronary Vascular Function and Structure). This descoped application is designed to address key aspects of the research goals put forward in Topic 6 of NASA Research Announcement (NRA) NNJ16ZSA001N-SRHHC “Space Radiation and Human Health Countermeasures Topics”: 1) to provide detailed physiological and bio-molecular characterization of risk and prediction for cardiac disease from microgravity and space-like radiation doses that are mission relevant for future human spaceflight beyond low Earth orbit and 2) to determine relative biological effectiveness (RBE) values for space radiation in appropriate animals using relevant intermediate as well as late physiological endpoints and effects. These experimentally derived RBE values will provide quantitative inputs into methods and models to calculate degenerative tissue responses in humans following exposures in space. The proposed studies center on assessment of the degenerative tissue effects of space radiation on cardiac disease. The central hypothesis of this project is that simulated weightlessness interacts with whole body exposure to space relevant doses of radiation to potentiate risk for cardiac disease. Three specific aims are proposed to directly address the issue of risk for cardiac disease using experimental approaches in a rat model relevant to the research questions. The proposed investigations directly address important aspects of the NASA Human Research Program (HRP) Degenerative Tissue Gaps, as set forth in the Human Research Roadmap. HRP gaps are identified and addressed by each Specific Aim. The proposed research will provide important information to help close these gaps. Specific Aim A: Simulate a 3-3.5 year deep space human mission in male WAG rats, determining the impact of simulated weightlessness combined with components of galactic cosmic rays using a simplified 5-ion mixed field beam (SimGCRSim) with total doses of 0.75 Gy and 1.50 Gy plus appropriate controls on risk for and occurrence of cardiac disease. Specific Aim B: Determine the response to gamma radiation in male Fischer 344 rats with and without simulated weightlessness to estimate relative biological effectiveness (RBE) for early clinically relevant endpoints of risk for and occurrence of cardiac disease. Rats will be irradiated with 3 doses of 137Cs: 3.0 Gy, 8.0 Gy, and 10.0 Gy at Brookhaven National Laboratory alongside experimental animals. Specific Aim C: Simulate a 3-3.5 year deep space human mission in male Fischer 344 rats, determining the impact of simulated weightlessness combined with components of galactic cosmic rays using a simplified 5-ion mixed field beam (SimGCRSim) with total doses of 0.75 Gy (provisionally) and 1.50 Gy plus appropriate controls on risk for and occurrence of cardiac disease.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research benefits life on Earth as it relates to patients who receive therapeutic radiation for the treatment of cancer and to individuals who receive accidental exposure to radiation from a nuclear accident.
Task Progress:	This research study addresses important aspects of the NASA Human Research Program Degenerative Tissue Gaps [Ed. note Dec 2020: Degenerative Risk was merged into CVD: Risk of Spaceflight Induced Cardiovascular Disease in July 2020; see https://]. Our studies provide physiological characterization for cardiac disease from microgravity and space-like radiation doses that are mission relevant for future human spaceflight beyond low Earth orbit. The interaction between simulated weightlessness and space radiation on cardiovascular disease needs to be understood prior to undertaking deep space exploratory missions. To date, combining the spaceflight hazard of reduced weight bearing via hind limb unloading (HLU) in rats of any strain with a simplified galactic cosmic ray simulation (SimGCRSim) in conscious animals has not been possible. The aim of our studies is to determine the impact of HLU combined with SimGCRSim using the fast beam switching and control systems technology recently developed at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory to determine occurrence of cardiac disease in a rat model. Here, we test the central hypothesis that simulated weightlessness interacts with whole body exposure to space relevant doses of radiation to potentiate risk for cardiac disease using a rat model with no pre-existing disease sensitive to radiogenic cardiac pathology. Ground-based studies were performed at the NSRL with male WAG/RijCmcr (Wistar) rats 6 months old at the start of the study. Rats were HLU for 5 days prior to, during, and 25 days after whole body irradiation with SimGCRSim with a total dose of 1.50 Gy. Other groups maintained normal weight bearing throughout the study. Other, non-HLU rats received a total dose of 1.5 Gy. Sham-irradiated and sham-HLU rats served as controls. The SimGCRSim is comprised of the following beams specified by NASA: protons (1Gev/n) – 35% of total dose, 28Si (600 MeV/n) – 1% of total dose, 4He (250 MeV/n) – 18% of total dose, 16O (350 MeV/n) – 6% of total dose, 56Fe (600 MeV/n) – 1% of total dose, protons (250 MeV/n) – 39% of total dose. One month post-irradiation, rats were transported to the Medical College of Wisconsin to determine long-term changes in risk for cardiac disease and kidney injury using blood samples withdrawn every 30 days over the 280 day follow up period. Perivascular collagen content in hearts were determined at the final time-point of this study at 280 days post-irradiation. Blood urea nitrogen (BUN), an index of kidney injury, was unchanged in rats irradiated with 1.5 Gy of SimGCRSim without HLU, when compared with sham-irradiated rats up to 280 days post-irradiation or sham-irradiation. BUN was unchanged in HLU rats compared with sham-irradiated HLU rats. No changes in BUN were observed in rats irradiated with 1.5 Gy of SimGCRSim, with or without HLU. Blood cholesterol, a risk factor for cardiac disease in humans, was unchanged in rats irradiated with 1.5 Gy of SimGCRSim compared with sham-irradiated rats. Cholesterol was also unchanged in HLU rats compared with sham-irradiated rats. Cholesterol was unchanged in rats irradiated with 1.5 Gy compared with rats that were HLU and irradiated with 1.5 Gy with normal weight bearing. However, cholesterol was transiently increased at 30, 120, and 210 days in rats that were HLU and irradiated with 1.5 Gy compared with rats that were HLU and sham-irradiated. Whole body exposure to 1.5 Gy SimGCRSim significantly increased (P = 0.024) cardiac perivascular collagen content, an index of cardiac fibrosis, compared with sham-irradiated, age-matched controls 280 days after irradiation or sham-irradiation. Notably, exposure of HLU rats to 1.5 Gy resulted in a significant increase (P = 0.008) of heart fibrosis

	compared with sham HLU, age-matched rats evaluated 280 days after irradiation or sham-irradiation. However, the addition of HLU to a 1.5 Gy dose of SimGCRSim that alone results in cardiac fibrosis did not change the degree of cardiac fibrosis beyond the effect of 1.5 Gy alone. The two groups could not be distinguished. Our findings provide evidence for the occurrence of cardiac fibrosis resulting from exposure of rats to a 5-ion mixed field beam designed to broadly simulate the composition of the GCR behind shielding, with no additive effect of the simulated spaceflight hazard of reduced weight bearing via HLU on the impact of SimGCRSim.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
Articles in Peer-reviewed Journals	Lenarczyk M, Laiakis EC, Mattson DL, Johnson BD, Kronenberg A, North PE, Komorowski R, Mäder M, Baker JE. "Irradiation of the kidneys causes pathologic remodeling in the nontargeted heart: A role for the immune system." FASEB BioAdvances. 2020 Dec;2(12):705-19. https://doi.org/10.1096/fba.2020-00071 , Dec-2020
Articles in Peer-reviewed Journals	Raber J, Fuentes Anaya A, Torres ERS, Lee J, Boutros S, Grygoryev D, Hammer A, Kasschau KD, Sharpton TJ, Turker MS, Kronenberg A. "Effects of six sequential charged particle beams on behavioral and cognitive performance in B6D2F1 female and male mice." Front Physiol. 2020 Aug 28;11:959. https://doi.org/10.3389/fphys.2020.00959 ; PMID: 32982769; PMCID: PMC7485338 , , Aug-2020